Md. Abdullah Al-Fahim

Civil Engineering Researcher | Computational Geomechanics

fahim.civilworks@gmail.com | linkedin.com/in/fahim-abdullah-f132
fahim9646.github.io | orcid.org/0009-0003-2483-4536
Google Scholar | Dhaka, Bangladesh

Research Summary

Civil engineer working between experimental soil mechanics and computational modelling. I build discrete element models of granular soils from scratch and calibrate them against matched laboratory tests, currently on how the inherent (bedding-plane) anisotropy of sand carries over to sand–structure interfaces. I am seeking a graduate (MSc/PhD) research position in computational geomechanics and related areas.

Interests: discrete element method; inherent anisotropy and fabric; soil–structure interfaces; contact-model calibration and validation; specimen-size effects in direct shear; sustainable construction.

Education

Rajshahi University of Engineering & Technology (RUET) 2019–24
BSc in Civil Engineering, Rajshahi, Bangladesh
CGPA: 3.06 out of 4.00

Research Experience

Undergraduate Thesis 2023 – Present
Dept. of Civil Engineering, RUET, Supervisor: Prof. Md. Mahmud Sazzad
"Influence of Inherent Anisotropy of Soil on the Behavior of Sand-Concrete Interface"

- Developed a custom two-dimensional, GPU-accelerated Discrete Element Method (DEM) engine from scratch in Python/PyTorch (float64) to simulate direct shear of clumped-grain assemblies, using elastic–perfectly plastic tangential and rolling-resistance contact laws and a sawtooth concrete interface.
- Designed a matched laboratory programme of 48 direct shear tests (ASTM D3080) on a river sand and a crushed sand, sheared against themselves and against concrete, at four bedding angles and three normal stresses, with three random-seed replicates per simulated condition.
- Applied a strict calibrate-once protocol: contact parameters tuned against experiment at a single bedding angle and then frozen, so that strength at all other bedding angles is a blind prediction rather than a re-fit.
- Analysed how inherent (bedding-plane) anisotropy is transmitted from sand–sand shearing to the sand–concrete interface, combining stress–strain response with contact-normal, force-weighted and grain-orientation fabric tensors.
- Studied the effect of specimen size on whether bulk anisotropy is captured in direct shear, and used the outcome to select the specimen geometry for the interface simulations.
- Next: extending the framework to sand–pile interfaces under lateral loading.

Barriers to Green Buildings in Bangladesh 2024
Survey study, RUET
Survey and Analytic Hierarchy Process analysis of the hindrances to green building construction ($n = 87$ responses; consistency ratio 0.011); government involvement ranked first (25.0%). First-author conference paper (ICCESD 2024).

Publications

Journal Articles

- Al-Fahim, M.A., Sazzad, M.M., Shafi, M.S.M. *Transmission and Attenuation of Bedding-Plane Anisotropy from Sand to Sand-Concrete Interfaces: A Discrete Element Study.* **In preparation.**

Conference Papers

- Al-Fahim, M.A., Shahriar, S. *Hindrances in Constructing Green Buildings in Bangladesh: A Comprehensive Study.* 7th International Conference on Civil Engineering for Sustainable Development (ICCESD 2024), KUET, Khulna, Bangladesh.

Technical Skills

- **Computational:** Python (PyTorch, custom DEM/FEM code), R, ABAQUS
- **Laboratory:** direct shear testing (ASTM D3080), particle-size analysis, concrete mix design (ACI 211)
- **Methods:** calibration and validation of numerical models, survey design, Analytic Hierarchy Process
- **Design & Productivity:** AutoCAD (2D), SketchUp, Lumion; MS Word, PowerPoint, Excel

Professional Experience

RANCON Engineering & Consultants Ltd 11/2025 - Present

Executive (QC-PMO), Tejgaon, Dhaka

- Inspect structural, architectural and finishing works and incoming materials against approved drawings, specifications, BOQ and BNBC; resolve non-conformities with contractors and keep QA/QC records (checklists, NCRs, corrective actions).
- Independently developed a live internal portfolio-tracking dashboard (Python/Flask) integrating project schedules with progress, dependency and financial reporting across concurrent developments.
- Developed and piloted RANCON QMS, a web-based digital inspection system: digital QC checklists, a raise–respond–verify–close NCR workflow, a role-based approval chain (Project Engineer, Project In-charge, HSE, QC/PMO), photo and PDF attachments, and in-app notifications.

Mir Cement Ltd. 01/2025 - 11/2025

Planning Engineer (Civil & RMC), Rupganj, Narayanganj

- Developed and monitored project schedules and prepared cost estimates, budgets and resource plans.
- Designed and optimised concrete mixes per ACI 211.R and verified raw materials and finished concrete against quality standards.

Presentations & Activities

- Poster Presentation — Retrofitting Old Toilets for Sustainable Bangladesh (**WaterAid Toilet Conference, 2025**)
- Team Event — Game of Trusses (**Ceenovation, IUT, 2024**)
- Poster Presentation — Sustainable Microgrids: Enhancing Campus Energy Resilience (**Technofair, RUET, 2023**)